Aakash Raman

Detection & Response Engineer | Threat Hunting | Incident Response | DFIR

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PROFESSIONAL SUMMARY

Detection & Response Engineer and DFIR enthusiast building MITRE ATT&CK-mapped detections as code in Sigma, YARA, and Python across endpoint, network, identity, cloud, and container telemetry. Experienced in incident response and malware analysis, and passionate about threat hunting, APT research, and finding evil in the environment. GIAC certified in GCFA, GNFA, GCFE, and GCIH.

EXPERIENCE

Associate Detection Engineer, Threat Hunt & Cyber Detection (THCD) | Morgan Stanley | Baltimore, MD Mar 2025 – Present

- Build MITRE ATT&CK-mapped detections, triage cases, and hunt for adversaries in a follow-the-sun model across endpoint (Windows, Linux, macOS), network and proxy (Netskope, Zeek), identity and SaaS (Microsoft 365 UAL, Entra ID), cloud, container (Wiz), and email telemetry using Sigma, YARA, Elasticsearch, Git, and Python.
- Conduct global threat hunts and adversary emulations targeting ransomware and APT groups, partnering with the Purple Team on SafeBreach simulations to find detection opportunities.
- Develop Sigma and Python cloud analytics for AWS, Azure, and GCP; onboarded Wiz for Kubernetes and Docker monitoring.
- Develop and tune analytics through a Detection-as-Code CI/CD pipeline built on GitLab and Prefect.
- Built Sigma and Python validators that enforce rule quality and keep detections aligned with the threat landscape, improving analytic validation coverage by 85%.
- Built automations with Parsec, an internal AI agent, to accelerate analytic tuning and script development.
- Conduct intelligence-driven hunts with the Threat Intelligence and Vulnerability Management teams, building detections and validating coverage for Threat Activity Alerts and emerging CVEs.
- Developed AI-assisted Python scripts for external threat hunting using urlscan.io, Recorded Future, and VirusTotal.
- Investigate business email compromise (BEC) cases for client fraud and block typosquatted domains impersonating the firm.
- Support the Incident Response team on network, endpoint, and cloud incidents using established DFIR playbooks.
- Perform static and dynamic malware analysis on suspicious executables and scripts using CFF Explorer, Ghidra, and dnSpy.
- Build Cortex XSOAR, CrowdStrike, and Defender playbooks to automate email purges, endpoint containment, and domain blocks.

Cybersecurity Analyst | Atlas Systems | East Brunswick, NJ (Remote) Aug 2024 – Mar 2025

- Built Azure security baselines (CIS v1.3) for a Rackspace-to-Azure migration, improving cloud security posture by 70% and closing 25% of identified gaps.
- Conducted vulnerability assessments with Nessus, Burp Suite, and Qualys, improving remediation by 30%, and automated PDF reporting in Python, reducing report generation time by 40%.

Cybersecurity Consultant Intern, X-Force Incident Response | IBM | Austin, TX May 2023 – Aug 2023

- Led purple team bootcamps with DFIR analysts using MITRE ATT&CK and Atomic Red Team, improving incident response by 95%.
- Tuned Splunk and QRadar detections and integrated STIX/TAXII threat intelligence feeds, improving detection efficiency by 40%.
- Reduced phishing incidents by 30% through threat hunts using OpenCTI, Cisco Talos, and YARA.
- Built insider threat playbooks in TheHive, Cortex, and CrowdStrike Falcon.

Cybersecurity Associate | Hevo Data Inc. | Bengaluru, India Jan 2021 – May 2022

- Reduced false positives by 30% by optimizing ELK SIEM queries and building custom alerts.
- Deployed DMARC and DKIM, reducing phishing and spam by 60%, and supported NIST CSF and ISO 27001 compliance audits.

CERTIFICATIONS

- **GIAC:** GCFA, GNFA, GCFE (2026); GCIH, GCCC, GSEC, GFACT (2025) | In progress: CISSP
- **Cloud:** AWS Certified Security – Specialty (2025), CCSK v5 (2025), AWS Solutions Architect – Associate (2023)
- **Other:** CompTIA CASP+, CySA+, PenTest+, Security+; CCD; Microsoft Applied Skills: Defender XDR; HTB Pro Lab: Zephyr

TOOLS & TECHNOLOGIES

- **Detection & SIEM:** Sigma, YARA, Elastic, Splunk, QRadar, Microsoft Sentinel, MITRE ATT&CK, Detection-as-Code
- **EDR & SOAR:** CrowdStrike Falcon, Microsoft Defender XDR, Cortex XSOAR
- **Cloud & Network:** AWS, Azure, GCP, Wiz, Kubernetes, Docker, Microsoft 365, Entra ID, Netskope, Zeek
- **DFIR & Malware Analysis:** Velociraptor, KAPE, Eric Zimmerman's Tools, Volatility, FTK Imager, Wireshark, Ghidra, dnSpy
- **Threat Intelligence & Emulation:** OpenCTI, TheHive, Recorded Future, SafeBreach, Atomic Red Team, Mythic C2
- **Programming Languages & Version Control:** Python, PowerShell, Bash, SQL, Git, GitLab CI/CD, Prefect

PROJECTS & OPEN SOURCE CONTRIBUTIONS

- **SigmaForage:** Python CLI that converts Sigma rules into native queries for 18+ SIEM and XDR platforms using pySigma backends.
- **SOF-ELK Contribution (PR #404, merged):** Added a parser for Eric Zimmerman's PECmd Prefetch output, resolving issue #163.
- **Visualizing Lateral Movement with Velociraptor JSON Logs:** Built Velocigrapher in Python, improving IR report quality by 85%.
- **Android Malware Analysis Research Paper:** Published in JESTR on Android malware behavior, C2 traffic, and mitigations.
- **Medium Blogs:** Articles on digital forensics, threat hunting, and detection engineering.

HIGHLIGHTS

- **Speaking:** BSidesCharm 2025 (Detecting Active Directory Attacks); OWASP Mumbai 2024 (Windows Malware Analysis).
- **CTFs:** 1st SANS x WiCyS 2024; 8th Amazon x WiCyS 2024; 34th of 4,267 OffSec Gauntlet 2025; CyberDefenders #32; TryHackMe top 1%.

EDUCATION

Master of Engineering, Cybersecurity | University of Maryland, College Park May 2024

B.Tech., Computer Science & Engineering (Information Security) | Vellore Institute of Technology Jul 2021